Fiscal Year:	FY 2023	**Task Last Updated:**	FY 12/27/2024
PI Name:	Sargsyan, Ashot M.D.		
Project Title:	Multipurpose Ultra-Portable Ultrasound Imager with Artificial Intelligence (AI) Capability		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	asargsyan@wylehou.com	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	
Organization Name:	KBR/NASA Johnson Space Center		
PI Address 1:	2400 NASA Pkwy Mail Code PMO		
PI Address 2:			
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	TRISH--Focused Investigations
Start Date:	06/01/2018	**End Date:**	09/30/2023
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 9/30/2023 per E. Urquieta/TRISH (Ed., 10/20/22) NOTE: End date changed to 9/30/2022 per E. Urquieta/TRISH (Ed., 8/3/21) NOTE: End date changed to 9/30/2021 per E. Urquieta/TRISH (Ed., 4/14/21) NOTE: End date changed to 02/28/2021 per TRISH (Ed., 8/20/20) NOTE: End date changed to 7/31/2019 per E. Urquieta/TRISH (Ed., 5/23/19)		
Key Personnel Changes/Previous PI:	Per closeout reporting from TRISH, Ronek Shah has been removed from the final list of Co-Investigators (Ed., 3/14/25).		
COI Name (Institution):	Ebert, Douglas Ph.D. (KBR/NASA Johnson Space Center) Byrne, Vicky M.S. (KBR/NASA Johnson Space Center) Pattarini, James M.D. (NASA Johnson Space Center)		
Grant/Contract No.:	NNX16AO69A-FIP0007		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Focused Investigation Project The goal of this project is to conduct ground-based and International Space Station (ISS) in-flight evaluation of new solid state ultrasound transducer technology with integrated artificial intelligence (AI) solutions in order to validate their operational usability in continuous microgravity and their innovation potential for operational medical support across all phases of ISS missions. This technology has high potential for valuable contributions to human health and performance capability maturation, in alignment with NASA's capability-based development goals. The investigation team will: 1) Select a high-Technology Readiness Level (high-TRL), Digital Imaging and Communications in Medicine (DICOM) compliant commercial off-the-shelf (COTS) ultrasound system, which incorporates new single-probe imaging technology, advanced data management with built-in AI solutions, intuitive user interface, and connectivity. The Butterfly iQ system (Butterfly Network Inc., N.Y., N.Y.), which is newly Food and Drug Administration-cleared (FDA-cleared) and will become available in 2018, is the only device on the market to meet all the above criteria. Most of its features are not available with other devices in the market. 2) Acquire iQ (2 units), other necessary equipment, and supplies for ground-based evaluation (Phase 1) 3) Perform a ground-based evaluation of iQ (Phase 1) a. Review hardware and electronics, interfaces, and operating functionality through analysis of manufacturer-provided specifications and testing data; b. Test the iQ (all units) using an approved quality assurance phantom, in parallel with a full-featured device for comparison purposes (GE VividQ or similar, i.e. ISS Ultrasound 2), including assessment of: contrast resolution at multiple depths; axial, lateral, and elevational resolution; penetration depth; c. Perform standardized multi-target imaging tests in human subjects by two trained sonography specialists (Ultrasound small and medium-sized enterprises (SMEs) in ten (10) human subjects (human-in-the-loop engineering evaluation with appropriate NASA approval). These tests will be duplicated with a full-featured system (similar to the current ISS Ultrasound 2) for comparison; Testing will be used to assess: • general usability and human factors, • image fidelity and limitations at three different depth ranges, • comparison of iQ with a full-featured system similar or identical to ISS Ultrasound 2. d. Perform standardized imaging tests and comparison with full-featured portable systems, by point of care ultrasound (POCUS) SME physician(s) using point-of-care ultrasound (POCUS) protocols (focused assessment with sonography in trauma (FAST) exam and lung ultrasound protocol) in 5 (five) human subjects (human-in-the-loop engineering evaluation, with appropriate NASA approval), and ultrasound-guided vascular access using a COTS ultrasound-guided vascular access phantom. e. Assess effectiveness of remote guidance using image streaming if available at the time of this evaluation. NDA and additional coordination with the Manufacturer for access to pre-release software may be required. If feasible, this will be performed by using novice operators in a limited scanning test in 5 (five) human subjects (human-in-the-loop engineering evaluation). f. Test effectiveness of AI solutions provided by the Manufacturer with novice operators using 10 human subjects (human-in-the-loop engineering evaluation) and observed by Ultrasound SMEs. 4) Perform analysis of formalized ratings, comments, and observations from all assessments and imaging tests (3 a-f above) by ultrasound SMEs with determination of appropriateness of iQ as a prototype for NASA human health and performance capability. 5) Prepare a Ground-Based Evaluation (Phase 1) Report. Presentation of results to TRISH and appropriate NASA organizations/boards.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This investigation examines multiple facets of novel ultrasound imaging technology, as well as its closer integration with delivery of medical care in various mission profiles and mission phases of human spaceflight. Scientific approach to device and technology evaluation, which includes comprehensive study of technical features, usability, data quality, human factors, and roles in various aspects of medical care, will result in well-substantiated recommendations. Multiple features of the device represent significant innovation. Of those, the single-probe solution based on Capacitive Micromachined Ultrasound Transducer (CMUT) with thousands of elements integrated with digital electronics in a highly compact design is highly attractive for a highly constrained spaceflight. Further, integration of deep-learning-based tools in the device software are a major step in the right direction of increasing crew autonomy and automated data processing. These and other features are closely examined by the team and will take a prominent place in the final report of the investigation. At this time, data analysis has not been completed, hence the information for this section remains limited and hypothetical.
	Novel technologies, approaches, and solutions implemented through the selected COTS ultrasound imaging device (Butterfly iQ, Butterfly Network Inc., Guilford, CT, USA) were evaluated for relevance to, and innovation potential for medical support of human space flight. In addition to evaluating the innovative imaging probe technology (Capacitive Micro-machined Ultrasound Transducer - CMUT) and its miniaturized implementation in a single-probe multipurpose device with a mobile app-based user interface, the project considered their effect in the operational context of human space flight, including remote guidance (ISS scenario) and autonomous use by inexperienced test subjects (Exploration scenarios). The device was also evaluated from the perspectives of acute care (emergency) procedures, as well as comprehensive diagnostic imaging protocols. Finally, the project demonstrated the feasibility and advantages of Artificial Intelligence / Machine learning - based tools in the context of space medical operational settings. In the second phase (extension, Phase 2) of the project co-fundd by TRISH and NASA, Butterfly iQ was flight certified and flown to the ISS for an engineering technology demonstration. AI/machine learning-based automated real-time tools (Auto-EF in ground testing, AutoBladder on ISS) were demonstrated and evaluated for human space flight purposes for the first time. The second phase of the project was conducted on ISS by means of human-in-the-loop engineering evaluation in complete crew autonomy, without prior knowledge or familiarization. For this purpose, the team leveraged its expertise in human factors and skill management to construct basic linear Just-ini-Time Instructions (JITI), which guided the crew to acquire images from multiple anatomical targets with adequate imaging quality and content. Thus, the

	combination of intuitive device operation and basic onboard instructions was shown as a valid approach for future autonomous imaging capability development.
Task Progress:	The results of this human-in-the-loop evaluation, while not producing human research data, already have a substantial effect on NASA's development of operational ultrasound imaging capability. Per the study team's recommendation, NASA adopted Butterfly iQ as a field medical tool to augment support in all hazardous training and operational sites on the ground. The demonstrated functionality and performance of this advanced single-probe handheld system in our ground study and in technology demonstrations (austere field conditions and the ISS testbed) resulted in the following recommendations to NASA; 1) To pursue steps required under NASA data security and cloud use policies for authorized use of device-native data management and connectivity solutions for both ground- and ISS – based use cases; 2) To pursue in-depth evaluation and ISS-based verification as a valid capability for specific operational uses (diagnosis, surveillance). This consideration would determine fitness of this device for augmenting the medical support capability, with tangible benefits such as near-instantaneous availability for medical contingency support, minimal use of infrastructure and personnel, streamlined data transfer, and transition to autonomous performance of some required surveillance procedures. 3) To advance technology demonstration and integration work onto exploration platforms. NOTE: After completion of this study's activities, NASA accepted the recommendation #2 above and, taking advantage of the available iQ payload on ISS, conducted a comprehensive scenario-based medical TD (Summer 2022).
Bibliography Type:	Description: (Last Updated: 05/19/2026)